

## Tutorial 7

Indian Institute of Technology Jodhpur  
Mathematics-I (MAL1010)

Instructors: Dr. Subhash Bhagat, Dr. Abu Talhamainuddin Ansary

---

1. Prove that the function  $f$  is defined on  $[a, b]$  by  $f(x) = x$ , is integrable on  $[a, b]$  by showing that  $\int_{-a}^b f = \int_a^{-b} f$ . Evaluate  $\int_a^b f$ .
2. A function defined on  $[0, 1]$  by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

Then show that  $f$  is not integrable on  $[0, 1]$ .

3. (a) Using only the definition of integral, show that

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right), & \text{if } x \text{ is irrational, } x \in [0, 1], \\ 0, & \text{otherwise, } x \in [0, 1]. \end{cases}$$

is not Riemann integrable.

- (b) Let  $f(x) = [x]$  for all  $x \in [0, 3]$ . Show that  $f$  is Riemann integrable and find  $\int_0^3 f(x) dx$ .
- (c) Let  $f(x) = e^x$  for all  $x \in [a, b]$ . Show that  $f$  is Riemann integrable and find  $\int_a^b e^x dx$ .
4. Let  $f : [a, b] \rightarrow \mathbb{R}$  and  $g : [a, b] \rightarrow \mathbb{R}$  be both integrable on  $[a, b]$  and there exist a positive real number  $k$  such that  $g(x) \geq k$  for all  $x \in [a, b]$ . Then  $f/g$  is integrable on  $[a, b]$ .
5. Let a function  $f : [a, b] \rightarrow \mathbb{R}$  be integrable on  $[a, b]$ . Then  $\int_a^b |f| \geq \left| \int_a^b f \right|$
6. A function  $f$  is continuous on  $[a, b]$  and  $f(x) \geq 0$  on  $[a, b]$ . If  $\int_a^b f = 0$ , prove that  $f = 0$  on  $[a, b]$  identically.
7. Evaluate the limit

$$\lim_{n \rightarrow \infty} \left( \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \cdots + \frac{1}{n+n} \right)$$

as an integral.